

# Artificial Intelligence in E-Entrepreneurship Training: Enhancing Digital Skills and Innovation Diffusion in Indonesia

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## ABSTRACT

This study explored the integration of artificial intelligence (AI) in e-entrepreneurship education, focusing on its impact on personalized learning, skill development, and student engagement. Using the diffusion of innovations framework, it examined factors influencing AI adoption, including infrastructure, educator preparedness, and ethical concerns. Through semi-structured interviews with Gen Z learners and focus groups with educators and AI technologists, the research identified key challenges—AI complexity, resistance to change, and data privacy. The study demonstrated that AI tools, such as personalized learning systems and business simulations, enhance entrepreneurial training by providing adaptive learning and real-time feedback. It concluded with policy recommendations to ensure inclusive access, especially for underserved entrepreneurs and contributes to understanding AI's transformative potential in bridging Indonesia's digital skills gap.

## KEYWORDS

Artificial Intelligence, E-Entrepreneurship, Indonesia, Education, Diffusion of Innovations

## INTRODUCTION

Artificial intelligence (AI) has become a crucial driver of digital transformation in global markets, particularly in e-commerce. AI applications, such as automated customer service, predictive analytics, personalized marketing, and fraud detection, have enabled businesses to operate more efficiently and adapt to dynamic consumer demands (Baijal et al., 2022). Indonesia, the largest digital economy in Southeast Asia, has experienced rapid expansion in e-commerce, fintech, and digital entrepreneurship. The *e-Conomy SEA 2024* report (Bain & Company et al., 2024) indicates that Indonesia's digital

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economy reached \$95 billion in U.S. dollars in gross merchandise value in 2024, with projections to exceed \$110 billion in U.S. dollars by 2025. The rise of AI-driven video commerce, which now accounts for 20% of Indonesia's total e-commerce gross merchandise value, represents a fundamental shift in digital shopping behavior, increasing from just 5% in 2022 (Bain & Company et al., 2023). Despite this strong growth, AI adoption remains uneven across businesses, especially among small and medium enterprises (SMEs). While large e-commerce platforms use AI-driven logistics and marketing, SMEs struggle with limited access to AI-powered digital marketing and financial analytics (Badan Pusat Statistik, 2023).

Indonesia's e-commerce sector accounted for 60% of the country's digital economy in 2024, growing at 15% each year (Bain & Company et al., 2024). AI-powered automation and digital marketing tools are revolutionizing customer engagement, supply chain management, and fraud detection (Baijal et al., 2022). One of the most significant AI-driven trends is video commerce, which integrates AI-powered recommendations and interactive shopping experiences. In Indonesia, video commerce gross merchandise value expanded from less than 5% in 2022 to 20% in 2024, reshaping consumer behavior and digital sales strategies (Bain & Company et al., 2024). However, AI adoption in Indonesia's e-commerce industry remains dominated by large marketplaces, such as Tokopedia, Shopee, and Bukalapak, while smaller businesses lack the resources to implement AI solutions (Badan Pusat Statistik, 2023). This highlights the need for affordable and accessible AI tools for SMEs.

The *e-Conomy SEA 2023* report highlights that Indonesia's digital payments market is projected to reach \$417 billion in U.S. dollars by 2025, driven by AI-powered payment systems and fraud detection algorithms (Bain & Company et al., 2023). AI is playing a pivotal role in streamlining financial transactions, assessing credit risk, and enabling personalized banking services (Baijal et al., 2022). Despite these advancements, AI-driven lending and wealth management remain underutilized among SMEs and micro-entrepreneurs, limiting access to financial resources for business growth (Bain & Company et al., 2023). Expanding AI-driven credit scoring models and automated lending platforms could enhance financial inclusion for digital entrepreneurs.

## Digital Skills Gap and Workforce Readiness

A critical challenge in Indonesia's e-commerce sector is the digital skills gap, particularly in AI expertise. According to the *Overview Indonesian Digital Society Index* report, Indonesia scored 58.25 out of 100 in the digital skills pillar, indicating widespread digital literacy, yet only 25.66 in the empowerment pillar, highlighting the disconnect between AI training and real-world business applications (Indonesian Digital Society Index, 2024). Similarly, the *Ipsos AI Monitor 2024: A 32-Country Ipsos Global Advisor Survey* (Ipsos, 2024) reported that 87% of Indonesians believe that AI will change their jobs within the next five years, but only 36% feel adequately prepared for these changes. This mismatch between AI awareness and AI skill development presents a significant barrier to innovation-driven entrepreneurship. Additionally, 69% of business leaders in Indonesia state that they would not hire individuals who lack AI competencies, reflecting the urgency of AI training initiatives (Badan Pusat Statistik, 2023).

Furthermore, AI literacy and usage remain highly concentrated in urban areas, while rural populations lag behind in AI adoption and e-commerce integration (Bain & Company et al., 2023). The Google-commissioned Kantar survey found that digital adoption rates among urban Indonesians are nearly double that of suburban users, particularly in e-commerce and fintech applications (Bain & Company et al., 2023).

## AI Training Initiatives and Policy Interventions

Indonesia has launched several digital skills development programs, yet AI-specific training remains limited. The Digital Talent Scholarship (DTS), managed by the Ministry of Communication and Digital, has provided digital literacy programs, but only 45.3% of Indonesian businesses currently offer AI-focused training to their employees (Indonesian Digital Society Index, 2024).

Additionally, most AI training programs cater to information technology professionals rather than entrepreneurs, leaving a significant gap in AI-driven business training (Badan Pusat Statistik, 2023). While Indonesia has invested heavily in AI infrastructure, the benefits are not yet evenly distributed among SMEs. The *e-Conomy SEA 2024* report stated that over \$30 billion in U.S. dollars was invested in AI infrastructure in Southeast Asia in the first half of 2024, but most of these investments have not directly supported AI adoption in SME entrepreneurship (Bain & Company et al., 2024). To bridge this gap, AI training programs should focus on natural language processing, AI-powered recommendation systems, and data analytics tailored for digital commerce (Baijal et al., 2022).

The rapid expansion of Indonesia's digital economy, particularly within the e-commerce sector, has highlighted the crucial role of AI in driving business innovation and operational efficiency. However, despite the increasing integration of AI-powered automation and digital marketing tools, there remains a significant skills gap among entrepreneurs, particularly in SMEs. The uneven distribution of AI adoption is further compounded by limited access to AI training, digital infrastructure disparities, and regulatory challenges. While government initiatives, such as the DTS, aim to improve digital literacy, these efforts have yet to focus explicitly on AI skill development for entrepreneurship (*WIFI DX Training: The Key to Digital Transformation for Indonesian Women Entrepreneurs*, 2024). Given these pressing issues, it is imperative to investigate how AI can enhance e-entrepreneurship training programs, the factors influencing AI adoption, and the challenges that educators face in implementing AI-driven tools.

## Research Questions

This study, therefore, seeks to answer the following research questions:

1. How can AI technologies improve the effectiveness of training and education programs in e-entrepreneurship?
2. What are the key factors influencing the adoption of AI-driven tools and platforms in e-entrepreneurship education, as explained by Rogers' (2003) diffusion of innovations (DoI) theory?
3. What are the key challenges e-entrepreneurship educators face when integrating AI-driven tools into their teaching methods?

To address these challenges, this study explores how AI technologies can improve e-entrepreneurship training by enhancing personalized learning, skill acquisition, and engagement. Furthermore, by applying Rogers' DoI theory, the research will analyze the key factors influencing the adoption of AI-driven tools and platforms in entrepreneurial education. Additionally, the study will identify the practical difficulties that e-entrepreneurship educators face when integrating AI-driven learning methods, such as resistance to change, lack of technical expertise, and infrastructure limitations.

## Research Objectives

The research objectives are structured to provide actionable insights into AI's role in entrepreneurship education and its broader implications for digital skills development in Indonesia:

- to explore the potential of AI technologies in enhancing personalized learning, skill development, and engagement in e-entrepreneurship education programs;
- to analyze the factors driving or hindering the adoption of AI-driven tools within educational contexts using the DoI framework; and
- to identify practical challenges that e-entrepreneurship educators face when integrating AI-driven tools into their teaching methods.

As Indonesia transitions into a more AI-driven digital economy, addressing the digital skills gap and AI literacy barriers in e-entrepreneurship training is paramount. While large enterprises increasingly leverage AI-powered business solutions, SMEs and digital entrepreneurs remain underserved in AI education and training programs. This disparity could hinder the long-term competitiveness and sustainability of Indonesia's digital economy. By investigating AI's potential in e-entrepreneurship training, identifying the factors driving its adoption, and addressing the challenges educators face, this study aimed to contribute to the development of an inclusive and innovation-driven digital workforce. The findings not only provide insights for policymakers and educators but also help bridge the digital divide between urban and rural entrepreneurs, fostering sustainable AI adoption across Indonesia's entrepreneurial ecosystem. A review of the geographical spread of academic literature shows a significant disproportion, as the majority of research is being produced in China, the United States, and the United Kingdom. Although there is some academic output from developing areas in Asia, Africa, and Oceania, the figures indicate that scholars in these regions must increase their contributions to the fields of AI and entrepreneurship (Boateng et al., 2024).

## **LITERATURE REVIEW**

E-entrepreneurship, characterized by the utilization of digital tools for enterprise creation and scaling, constitutes a significant transformative force in the contemporary global economy. Traditional educational models frequently do not satisfy the evolving requirements of this field, despite their potential. The incorporation of AI in e-entrepreneurship education effectively addresses this gap, providing innovative solutions that correspond to the swiftly changing digital environment. Research indicates that AI-driven tools can offer personalized guidance, optimize skill development, and improve the scalability of educational programs (Labyad et al., 2024). AI has become a significant influence in education, providing tailored support, optimizing skill acquisition, and improving the scalability of educational initiatives. The incorporation of AI-driven tools into educational systems has demonstrated the ability to address individual learning requirements, thus promoting a more adaptive and efficient learning environment. AI tools facilitate personalized learning experiences by customizing educational content to address the individual needs of each learner. Research shows that AI platforms provide tailored learning materials, immediate feedback, and self-assessment opportunities, effectively addressing individual strengths and weaknesses while fostering metacognitive and self-regulatory skills in learners (Wei, 2023). This personalized approach represents a notable shift away from conventional educational models that typically employ a uniform strategy, thereby improving student engagement and motivation (Abbas et al., 2023).

Another benefit of AI incorporation is educational program scalability. AI-driven e-learning systems can make education more accessible by offering continuous and economical learning (Murtaza et al., 2022). These systems can adjust material to learners' learning trajectories, maximizing educational resource use (Murtaza et al., 2022). AI solutions can also help instructors manage administrative work, letting them focus on teaching (Aghaziarati et al., 2023). This simultaneous focus on pedagogical and administrative efficiency highlights AI's diverse effects on education (Aghaziarati et al., 2023). AI in education goes beyond content delivery to improve learning outcomes through predictive and learning analytics. AI systems can anticipate student learning outcomes and inform instructional techniques by analyzing student performance data (Ouyang et al., 2023). This helps educators to identify at-risk pupils and create focused interventions that improve educational results (Ambarita & Nurahmatullah, 2024). Rogers' (2003) DoI theory explains how social systems spread breakthroughs, like AI in education. This theory states that innovation adoption is influenced by innovation, communication channels, timing, and social systems (Fuah & Ganggi, 2022; Ismail, 2023).

## AI in Pedagogical Practice

AI is changing entrepreneurial education, especially in developing crucial skills. AI technology, such as adaptive learning systems, simulations, and predictive analytics, enable tailored learning that meets entrepreneurship program goals. These technologies boost engagement and help develop entrepreneurial abilities. Zhao (2021) highlighted AI-powered virtual reality platforms that make complicated entrepreneurship challenges more approachable and entertaining for students. This coincides with Rogers' (2003) factors of relative advantage and compatibility because these AI technologies allow students to experiment with entrepreneurial ideas risk-free.

According to Somià and Vecchiarini (2024), generative AI can encourage learners to think creatively and critically. This is vital in entrepreneurship education, as innovation drives success. Mahmudin (2023) found that entrepreneurship education not only teaches theory but also develops skills, like creativity, leadership, and strategic thinking, which are crucial for navigating a rapidly changing economy. Kakouris (2021) also emphasized experiential learning and critical reflection in entrepreneurial education, claiming that AI can provide real-time feedback and adaptable learning pathways.

## Factors Influencing AI Adoption in Education

To understand the key factors influencing the adoption of AI-driven tools and platforms in e-entrepreneurship education through the lens of Rogers' (2003) DoI theory, we can draw insights from several relevant studies. These factors are categorized based on Rogers' five attributes.

### *Relative Advantage*

AI products' apparent benefits—better learning and corporate performance—are vital. SMEs embrace AI in e-commerce to increase business performance (Aljarboa, 2024; Horani et al., 2023). AI adoption increases if organizations see it as cost-effective (Horani et al., 2023). Benefits and cost-effectiveness drive AI tool adoption in SMEs and healthcare. Businesses' readiness to adopt AI depends on these criteria.

AI technologies are adopted in this area due to perceived benefits, including increased productivity and operational efficiency (Istiqomah & Alfansi, 2023; Santosa & Surgawati, 2024). AI has also been shown to improve diagnostic accuracy and workflow efficiency in healthcare (Alnasser, 2023; Jiao et al., 2023). AI applications in ophthalmology have improved screening processes, improving patient outcomes and diagnostic times (Lin et al., 2022; Pietris et al., 2022).

### *Compatibility*

AI tools must fit existing systems and processes. Technical compatibility strongly determines e-learning and AI adoption (Chan & Ngai, 2012; Horani et al., 2023). AI adoption in education depends on technical compatibility and strategic synergy. Technical compatibility refers to how effectively AI technologies fit into existing systems and procedures. Research shows that compatibility substantially impacts e-learning and AI adoption. Wong and Yap's study found that technical compatibility had a statistically significant effect on AI adoption, with a  $p$ -value of 0.02, suggesting that organizations are more likely to adopt AI if they think that it fits their practices and standards (Wong & Yap, 2024). Seethamraju and Hecimovic (2022) agree that AI technologies must be compatible with audit and accounting systems to be effective.

### *Complexity*

The perceived ease of use of AI tools is crucial. User-friendly tools with low technical requirements are more likely to be used (Haces et al., 2024; Horani et al., 2023). AI tools' perceived ease of use influences their uptake across sectors, especially education and healthcare. According to recent studies on the role of intuitive design in facilitating acceptance and integration into

workflows, users are more likely to adopt user-friendly solutions that require less technical understanding. Geddam et al. (2024) noted that user-friendly interface design boosts user engagement and pleasure, increasing adoption.

### *Trialability*

Knowing how healthcare institutions can adopt new technologies requires knowing trialability in AI implementation. Pilot programs allow institutions to test AI tools on a smaller scale and assess their pros and cons before implementing them. Horani et al. (2023) noted that pilot programs allow a controlled assessment of AI tools, improving understanding of their operational dynamics and healthcare delivery effects. Pilot programs work in medical applications, according to recent research. Poli et al. (2024) showed that an AI cervical cancer screening program may achieve 76% diagnostic accuracy in real life.

### *Observability*

Visible advances in education or corporate performance can spur adoption. AI-driven e-learning platforms that improve student engagement and learning outcomes may stimulate greater use (Haces et al., 2024; Horani et al., 2023). Observability is essential for educational and commercial uptake and outcome improvement. Innovative methods, like problem-based learning and technology-enhanced learning, can improve student engagement and learning results in schools. Rahman et al. (2024) asserted that integrating problem-based learning with augmented reality creates immersive environments that enhance student engagement and economic understanding. Berry et al.'s (2020) systematic analysis found that technology-enhanced learning improves periodontics education, suggesting that comparable methods could benefit other educational disciplines.

## **KEY CHALLENGES IN INTEGRATING AI IN E-ENTREPRENEURSHIP EDUCATION**

Integrating AI-driven tools into e-entrepreneurship education presents several challenges, including pedagogical, technical, and ethical concerns.

### **Pedagogical Challenges**

Due to the rapid advancement of AI technologies, finding effective tools and best practices is difficult (Labyad et al., 2024). Competitive forces and regulatory obligations are driving AI tool adoption in enterprises. AI technologies are needed to improve operational efficiency and meet changing industry norms. AI is being used in human resource management to expedite recruiting and improve decision-making.

AI has been found to improve human resource management talent recruiting efficiency. AI-supported selection technologies, like résumé screening and interview systems, help organizations to process applications faster and more consistently (Hunkenschroer & Luetge, 2022; Köchling & Wehner, 2022).

### **Technical Issues**

Another major obstacle to AI inclusion in education is instructors' technological inexperience. Professional development and training programs are needed, since many instructors lack AI tool skills (Ng et al., 2023). To keep up with AI technology's rapid evolution, educators must practice new skills (Babashahi et al., 2024). Integration is complicated by high infrastructure and maintenance expenses (Ala et al., 2022). Technical hurdles arise from resource inequities, instructor inexperience, and expensive infrastructure expenses when integrating AI into education. These obstacles prevent educational institutions from using AI technologies fairly and effectively. Resource differences



across universities are a major issue affecting AI access. According to Edeni et al. (2024), urban and rural areas have different amounts of digital infrastructure, hardware, and software. Aghaziarati et al. (2023) noted that many educators are unprepared to use AI in their classrooms, so they require continual training. According to Ferikoğlu and Akgün (2022), teachers' awareness of AI and views toward its use in education are correlated.

## Issues of Ethics and Practice

Overusing AI tools can impair critical thinking and problem-solving (Zhou et al., 2024). Research shows that people develop heuristics that favor AI suggestions, even when they are wrong (Buçinca et al., 2021). Integrating AI into numerous areas, especially education, poses ethical and practical challenges that must be considered. User overreliance on AI tools can impair critical thinking and problem-solving. Over-reliance can lead to an inability to evaluate information critically, which hinders autonomous problem-solving (Cao et al., 2024). Integrating AI into decision-making requires careful consideration of ethical concerns, notably academic integrity and data protection. AI tools in education raise worries about academic honesty and disinformation (Wan et al., 2024).

## RESEARCH METHODOLOGY

### Qualitative Approach

This study employed a qualitative research design to investigate how Rogers' (2003) DoI theory manifests in the adoption of AI within e-entrepreneurship education. A qualitative approach is particularly suited for this research because it allows for an in-depth exploration of the perceptions, experiences, and decision-making processes that influence AI adoption in educational and entrepreneurial contexts (Creswell & Poth, 2018).

By utilizing focus group discussions (FGDs) and semi-structured interviews, this study examined educators' and technologists' perspectives on AI adoption, while incorporating insights from Generation Z learners who have participated in DTS programs relevant to AI applications in e-entrepreneurship training.

### Data Collection Methods

#### *FGDs*

This study involved 13 key informants, including educators and AI engineers, to examine AI integration in e-entrepreneurship education. FGDs explored pedagogical strategies, AI-driven tools, and best practices for optimizing instructional effectiveness. Beyond pedagogy, discussions addressed challenges and opportunities in AI adoption, including infrastructure constraints, resistance to change, and educator training needs. Participants also highlighted AI benefits, such as personalized learning, real-time feedback, and experiential simulations. These insights contribute to a comprehensive understanding of AI's role in e-entrepreneurship education. FGDs utilized open-ended questions to encourage in-depth discussions on AI implementation, effectiveness, and barriers (Krueger & Casey, 2015). Sessions included 3–4 participants, lasting 60–90 minutes, with audio recordings and verbatim transcriptions for analysis. Participants reviewed summaries for verification, enhancing credibility (Lincoln & Guba, 1985).

#### *Semi-Structured Interviews*

This study also conducted semi-structured interviews with 20 Generation Z learners from DTS programs on AI in e-entrepreneurship. These interviews captured firsthand experiences and assessed AI's impact on learning, skill development, and business innovation. Discussions explored AI's effectiveness, entrepreneurial benefits, and alignment with educational goals (Brinkmann, 2020). Interviews were conducted face-to-face, virtually, or in hybrid formats, lasting 30–45 minutes. All

interviews were transcribed, anonymized, and thematically analyzed for a rigorous and systematic examination of insights. The discussion questions for the interviews and FGDs are shown in Table 1.

Table 1. Interview and Focus Group Discussion (FGD) Question Guide

| Theme                                              | Questions                                                                                                     |
|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| AI integration in e-entrepreneurship education     | 1. How have you integrated AI technologies into e-entrepreneurship education?                                 |
|                                                    | 2. What benefits have you observed from using AI in the teaching process?                                     |
|                                                    | 3. What challenges have you encountered in implementing AI tools in education?                                |
| Factors influencing AI adoption                    | 1. What factors do you believe facilitate the adoption of AI in e-entrepreneurship education?                 |
|                                                    | 2. What are the key barriers to AI adoption within your institution?                                          |
|                                                    | 3. How compatible are AI tools with the current educational systems and curricula in your institution?        |
| Support and professional development for educators | 1. What type of training or support is needed for educators to effectively utilize AI tools?                  |
|                                                    | How prepared do you think educators are in adopting and integrating AI in their teaching practices?           |
| Ethical concerns and data privacy                  | 1. What ethical concerns have emerged regarding the use of AI in educational settings?                        |
|                                                    | 2. What measures should institutions take to address data privacy and security issues related to AI adoption? |

*Note.* This table provides a clear and structured overview of the questions used during the interview and focus group discussion (FGD) sessions. The questions are organized by theme to align with the study's objectives and focus areas. AI = artificial intelligence.

## Data Analysis

### *Thematic Analysis*

This study applied thematic analysis to identify key themes in AI adoption for e-entrepreneurship education, aligning with Rogers' (2003) innovation diffusion model (Braun & Clarke, 2006; Dunan et al., 2025). NVivo software supported data organization, coding, and pattern recognition, ensuring systematic and reliable qualitative analysis (Silver & Lewins, 2014).

### *Data Triangulation for Credibility*

Data triangulation enhanced validity by cross-verifying insights from FGDs, interviews, policy documents, case studies, and academic literature. This approach identifies consistent and divergent themes, strengthening analytical rigor. Patton (1999) highlighted triangulation's role in integrating multiple perspectives for more credible findings.

### *Ethical Considerations*

This study upheld strict ethical standards. Data collection followed ethics board approval, ensuring participant rights and confidentiality. Informed consent preceded recordings, clearly outlining research objectives and participant rights. Identities remained anonymous, and participation was voluntary, allowing the right to withdraw anytime. Secure data storage ensured confidentiality and research integrity.



## RESULTS AND DISCUSSION

### AI Integration in E-Entrepreneurship Education

The integration of AI in e-entrepreneurship education is a critical research domain with substantial potential to enhance learning outcomes and entrepreneurial competencies. This study applied thematic analysis to semi-structured interviews with 20 Generation Z learners and FGDs with educators and AI specialists, providing a comprehensive examination of AI-driven tools in digital entrepreneurship education.

The findings emphasize AI's transformative impact on personalized learning, skill development, and student engagement within entrepreneurial training programs. Additionally, this study investigated key factors influencing AI adoption, pedagogical challenges that educators face, and AI's evolving role in shaping e-entrepreneurship education. The results have been categorized into three core themes, offering a structured perspective on AI's contribution to entrepreneurship learning.

To contextualize the participants' backgrounds, Table 2 presents detailed demographic and educational profiles of interviewees and FGD participants. This dataset provides valuable insights into the diverse experiences and levels of engagement of Generation Z learners, educators, and AI technologists, illustrating their varying degrees of familiarity with AI-driven educational tools in the digital entrepreneurship ecosystem.

**Table 2. Characteristics of Informants for Interviews and Focus Group Discussions (FGDs)**

| Type of Informant                         | Age Range   | Generation   | Education Level                         | Number of Informants | FGD Group Size             |
|-------------------------------------------|-------------|--------------|-----------------------------------------|----------------------|----------------------------|
| Semi-structured interviews (Generation Z) | 19–26 years | Generation Z | High school (7), bachelor's degree (13) | 20                   | N/A                        |
| Educators and technologists (FGD)         | 30–40 years | Millennials  | PhD                                     | 3                    | 3–4 participants per group |
|                                           | 41–56 years | Generation X | PhD                                     | 4                    | 3–4 participants per group |
|                                           | 57+ years   | Baby boomers | PhD                                     | 4                    | 3–4 participants per group |

*Note.* FGD = focus group discussion; PhD = Doctor of Philosophy.

Table 3, on the other hand, provides a detailed overview of the themes and sub-themes that emerged from the data analysis. These themes highlight the various aspects of AI integration in e-entrepreneurship education, such as personalized learning approaches, AI adoption barriers, and institutional readiness. By categorizing the results into these themes, Table 3 sets the stage for a more in-depth discussion of the findings and their implications for AI-enhanced entrepreneurship training.

**Table 3. Themes and Sub-Themes**

| Theme                                                                  | Sub-Themes                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. AI-driven enhancements in e-entrepreneurship training and education | <ol style="list-style-type: none"> <li>1. Personalized learning and adaptive AI-based instruction</li> <li>2. AI-powered skill development and experiential learning</li> <li>3. Gamification and AI-based engagement in learning</li> <li>4. AI and business idea validation in e-entrepreneurship</li> <li>5. AI for business networking and market access</li> </ol> |
| 2. Factors influencing ai adoption in e-entrepreneurship education     | <ol style="list-style-type: none"> <li>1. Relative advantage</li> <li>2. Compatibility</li> <li>3. Complexity</li> <li>4. Trialability</li> <li>5. Observability</li> <li>6. External influences and institutional support</li> </ol>                                                                                                                                   |
| 3. Challenges faced by educators in AI integration                     | <ol style="list-style-type: none"> <li>1. Technical and infrastructure barriers</li> <li>2. Educator readiness and digital literacy</li> <li>3. Data privacy, ethics, and bias in AI</li> <li>4. Student adaptability and acceptance of AI in learning</li> <li>5. Institutional policy and regulatory gaps</li> </ol>                                                  |
| 4. Future directions for AI in e-entrepreneurship education            | <ol style="list-style-type: none"> <li>1. Best practices for AI adoption in e-entrepreneurship training</li> <li>2. Policy recommendations and institutional AI strategies</li> <li>3. Scaling AI for inclusive digital entrepreneurship</li> </ol>                                                                                                                     |

*Note.* AI = artificial intelligence.

### Theme 1: AI-Driven Enhancements in E-Entrepreneurship Training and Education

The integration of AI into e-entrepreneurship education offers transformative opportunities that enhance both the learning process and the effectiveness of entrepreneurship training. AI enables personalized learning, improves skill development, and provides innovative methods for validating business ideas, which are essential for nurturing future entrepreneurs.

AI-driven platforms offer personalized learning by adapting to individual student needs. Informant 7, a Generation Z interviewee with a bachelor’s degree, noted that AI systems allow students to progress at their own pace, adjusting content difficulty based on their performance. This adaptability is especially valuable in the e-entrepreneurship education context in which students’ digital literacy varies. Informant 5 highlighted how AI customizes content for different levels of expertise, providing additional resources for those struggling with complex concepts, ensuring inclusivity within diverse learner profiles.

Beyond personalization, AI significantly contributes to practical skill development through experiential learning tools, like simulations and virtual reality. Educator 2, a millennial FGD participant, emphasized that AI-powered platforms allow students to experiment with business strategies in a risk-free environment, refining their decision-making and problem-solving skills. Similarly, technologist 4, a baby boomer, observed that AI simulations of market conditions and customer interactions provide students with a deeper understanding of the impact of their decisions before applying them in real business contexts.

AI’s application in gamification further enhances student engagement. Informant 9, a bachelor’s degree student, shared how AI-driven gamified platforms increases their interest in e-entrepreneurship by rewarding achievements, such as completing lessons and business simulations. These gamification

elements keep students motivated and engaged, which not only boosts retention but also immerses them in real-world business challenges, enriching their learning experience.

AI also plays a crucial role in business idea validation. Educator 1 explained that AI tools enable students to test their business ideas against historical market data, consumer trends, and competitor analysis, providing invaluable insights into the feasibility of their concepts. This data-driven approach helps students to refine their business models and identify potential risks early in the entrepreneurial process, enhancing their chances of success.

Additionally, AI facilitates business networking and market access. Technologist 2 highlighted how AI algorithms help students to connect with industry experts, potential partners, and investors, offering tailored opportunities that align with their business interests. Informant 15 reflected on how AI tools assist in market analysis, enabling students to access global markets and gain a deeper understanding of target audiences, which significantly enhances their strategic business planning.

AI-driven advancements in e-entrepreneurship education are revolutionizing learning by making it more personalized, engaging, and practically focused. These innovations not only equip students with essential entrepreneurial skills but also provide the tools necessary to validate their ideas and refine their strategies, ultimately increasing their likelihood of success in a competitive digital economy.

## **Theme 2: Factors Influencing AI Adoption in E-Entrepreneurship Education**

The adoption of AI in e-entrepreneurship education is shaped by a range of factors that either facilitate or hinder its integration into learning environments. From the perspectives of both educators and students, several key elements are critical in determining the success of AI adoption, including its relative advantage, compatibility with existing systems, complexity, trialability, observability, and external influences.

The relative advantage of AI over traditional teaching methods is a primary motivator for its adoption. Educator 3, a millennial participant, highlighted AI's ability to automate repetitive tasks and provide real-time feedback, offering a more efficient and dynamic learning experience. Informant 12, a student, noted that AI tools quickly identify learning gaps and provide immediate support, which traditional methods often fail to deliver. This perceived advantage fosters a more responsive learning environment that adapts to individual needs, making it an attractive tool for both students and educators.

Compatibility with existing curricula and pedagogical frameworks is crucial for AI tools to gain acceptance. Technologist 1, a Generation X informant, identified compatibility as a significant challenge, particularly in institutions that have yet to embrace digital transformation. However, educator 5 observed that when AI tools complement existing teaching strategies rather than replace them, they are more readily accepted. For instance, AI can enhance project-based learning by automating performance analysis without disrupting the overall educational approach, facilitating smoother integration into the curriculum.

Complexity is another critical factor in AI adoption. Informant 3, a high school graduate, pointed out that the perceived complexity of AI tools can be a barrier, especially for those with limited technical expertise. Technologist 3, a baby boomer FGD participant, emphasized the importance of creating intuitive and user-friendly AI systems. Simplified interfaces and targeted training for both students and educators are essential to overcoming this challenge and ensuring the effective use of AI in educational settings.

Trialability, or the opportunity to experiment with AI tools before full implementation, is another key determinant of successful adoption. Educator 2 explained that pilot testing allows both educators and students to explore AI tools in a low-risk environment, enabling feedback and adjustments which can improve integration strategies. Informant 10, a student, shared how experimenting with AI systems in business simulations has deepened their understanding of how AI can support decision-making processes in entrepreneurship.

Observability of AI's impact on learning outcomes is crucial for sustaining its adoption. Technologist 4 emphasized the importance of tracking improvements in student engagement,

performance, and business idea validation. Informant 18 noted that AI-powered feedback and progress tracking allowed them to monitor their growth, refining their approach to business development. When students can directly observe the benefits of AI tools in their learning journey, they are more likely to embrace and continue using these technologies.

External influences, such as government policies, industry trends, and institutional support, also play a significant role in AI adoption. Educator 4 noted that government initiatives promoting digital skills training have facilitated the integration of AI tools into educational curricula. Institutional support in the form of funding, infrastructure, and professional development is equally crucial. Informant 16 emphasized that their university's commitment to providing AI-based resources and training has enabled effective use of AI tools in entrepreneurship courses, underscoring the importance of institutional backing in fostering successful AI adoption.

The factors influencing AI adoption in e-entrepreneurship education are interconnected and multifaceted. To ensure the successful integration of AI, institutions must address these factors, particularly the relative advantages of AI, its compatibility with existing educational frameworks, and the complexity of the tools themselves. By considering these elements, institutions can create an environment in which AI technologies enhance the learning experience, better prepare students for entrepreneurial challenges, and increase the overall effectiveness of digital entrepreneurship education.

### **Theme 3: Challenges Faced by Educators in AI Integration**

The integration of AI into e-entrepreneurship education presents several challenges which need to be addressed for successful implementation. These challenges span technical, pedagogical, ethical, infrastructural, and institutional dimensions. Understanding and overcoming these barriers is essential to creating an environment in which AI tools can effectively enhance teaching and learning in entrepreneurial training programs.

A significant obstacle to AI integration is the lack of technical infrastructure. As educator 6, a Generation X participant, pointed out, many institutions, particularly in developing regions, struggle with outdated technology, poor internet connectivity, and insufficient hardware. These limitations prevent the full implementation of AI-based tools, which require advanced computing power and stable online platforms. Technologist 2 further emphasized that to adopt AI effectively, institutions must invest in modern infrastructure, including high-speed internet and cloud computing services, alongside providing adequate training for both educators and students.

In addition to technical barriers, educators' readiness to adopt AI presents a substantial challenge. Informant 4, a student with a high school education, noted that many instructors continue to rely on traditional methods and lack the skills necessary to incorporate AI into their teaching. Educator 3 echoed this concern, stressing the importance of professional development programs that equip educators with the skills to use AI tools effectively. However, technologist 5 added that the challenge extends beyond training; it involves overcoming a mindset that views traditional methods as sufficient for modern education. Creating a culture of continuous learning and innovation among educators is critical for successful AI integration.

Ethical concerns, particularly regarding data privacy and algorithmic bias, are also major barriers to AI adoption. Informant 19 highlighted the risks associated with AI systems that collect personal data, as these datasets are often processed by third-party vendors, raising concerns over security and ownership. Moreover, AI algorithms may perpetuate biases if the data on which they are trained is incomplete or unrepresentative. As educator 2 pointed out, biased data leads to biased outcomes, which can disproportionately affect students from marginalized groups. Addressing these ethical issues requires robust data privacy protocols, transparency in the use of AI tools, and AI systems that are trained on diverse, unbiased datasets. Clear ethical guidelines are also essential to ensuring fairness and privacy in AI-driven education.

Student adaptability and acceptance of AI technologies also pose challenges. Informant 8 noted that some students may struggle to adopt AI-based tools, especially if they are not familiar with

digital technologies. Informant 7 pointed out that, while AI offers significant benefits, students may be reluctant to trust AI-driven assessments or recommendations, preferring the human touch in feedback. Overcoming this resistance requires AI systems that are intuitive, transparent, and user-friendly. Educators must play a crucial role in guiding students through these technologies, explaining their benefits and demonstrating how they can enhance learning. Educator 4 suggested that gradual implementation of AI tools would allow students to become familiar with their use and appreciate their value before full integration into the curriculum.

Institutional policies and regulatory frameworks also present challenges for AI integration. Educator 1 emphasized the need for institutions to update their policies to support AI use, particularly in areas, such as curriculum design, data privacy, and teacher training. Without clear, coordinated policies, AI adoption may be inconsistent, leading to inefficiencies. The evolving regulatory landscape surrounding AI in education further complicates matters. As technologist 3 suggested, policymakers must collaborate with educational leaders to create frameworks that not only support the ethical use of AI but also ensure equitable access for all students, regardless of their socioeconomic status.

While the integration of AI into e-entrepreneurship education offers significant promise, overcoming the technical, pedagogical, ethical, and institutional challenges is essential. By addressing issues, such as infrastructure, educator readiness, data privacy, student acceptance, and policy alignment, educational institutions can unlock AI's full potential to revolutionize entrepreneurship education and better prepare students for success in the rapidly evolving digital economy.

#### **Theme 4: Future Directions for AI in E-Entrepreneurship Education**

The future of AI in e-entrepreneurship education is marked by significant growth potential, driven by emerging technologies and innovative pedagogical approaches that promise to transform learning environments. As AI evolves, its role in shaping the future of entrepreneurial education will become increasingly indispensable. This section examines key strategies for successful AI adoption, policy recommendations, and ways to scale AI for inclusive digital entrepreneurship, which can foster innovation and broaden opportunities for entrepreneurs worldwide.

Effective AI adoption in e-entrepreneurship education requires adhering to best practices. Educator 6 emphasized that institutions should start with a clear understanding of AI's potential value, adopting a phased approach that begins with pilot programs and scales based on feedback and measurable results. Informant 13 further noted the importance of continuous professional development for educators, ensuring that they are equipped with both technical and pedagogical expertise to leverage AI effectively. Technologist 5 argued that AI adoption should focus on scalability, ensuring that AI tools are adaptable to diverse educational settings and provide robust returns on investment. Incorporating AI tools that track student progress in real-time can further enhance instructional practices, making teaching more responsive and data driven.

Policy recommendations are crucial for fostering a supportive environment for AI adoption. Educator 2 advocated for government-backed initiatives that provide funding and resources to help educational institutions invest in AI technologies. Public-private partnerships can bridge the gap by offering affordable AI solutions, particularly for SMEs and educational institutions with limited resources. Technologist 3 stressed that policies should protect student data privacy while ensuring that AI enhances the learning experience. Transparent AI systems that are free from bias will foster trust and promote broader adoption. Informant 17 highlighted that the future of AI in education hinges on balancing innovation with responsible governance, ensuring that AI is used ethically, while safeguarding individual rights.

Scaling AI for inclusive digital entrepreneurship presents a promising opportunity to democratize access to entrepreneurial education. Informant 9 observed that AI technologies can be scaled to support students in remote or underserved areas, where traditional educational resources are often lacking. AI-powered platforms can provide essential tools, guidance, and feedback, enabling students to develop business ideas without requiring physical classrooms. This scalability opens up

opportunities for a broader audience, including marginalized and rural populations, democratizing access to entrepreneurship education. Educator 4 called for the creation of AI-based business incubators to support aspiring entrepreneurs with mentorship and resources, while technologist 1 noted that AI's ability to analyze large datasets and identify market trends can empower entrepreneurs in developing countries, giving them a competitive edge.

AI is also positioned to drive inclusive innovation within the digital entrepreneurship ecosystem. Technologist 4 emphasized that AI can help to identify market gaps and facilitate the creation of innovative solutions. By analyzing market data, entrepreneurs can quickly adapt to shifts in consumer preferences and technological advances, which is critical for success in a rapidly changing global market. Educator 5 highlighted that AI enables students to experiment with new business models, allowing them to test and validate ideas with minimal risk and cost. As AI becomes more accessible, it will empower a new generation of entrepreneurs who can leverage digital tools to tackle real-world challenges. Informant 2 added that AI tools offer unprecedented opportunities for innovation, enabling students to build scalable, sustainable businesses with a global reach.

Addressing the ethical challenges that AI poses will be essential for its future integration in e-entrepreneurship education. Educator 1 cautioned that, while AI holds immense potential, its widespread use must be carefully managed to avoid misuse or harm. Clear policies must be established to ensure that AI systems are transparent, fair, and aligned with educational goals. AI algorithms should be designed to reflect diverse perspectives and avoid reinforcing societal biases, ensuring that AI contributes to inclusive innovation rather than exacerbating inequalities. Technologist 2 reiterated the importance of transparency, stressing that AI tools should be developed with clear accountability and ethical standards to maintain trust in educational environments.

The future of AI in e-entrepreneurship education offers exciting possibilities that can drive innovation, enhance educational outcomes, and equip students with the skills necessary for success in the global digital economy. By following best practices for AI adoption, establishing robust policies, and ensuring AI accessibility, institutions can create a dynamic and inclusive entrepreneurial ecosystem. These advancements will not only benefit students but also contribute to the growth of small businesses and the broader economy, fostering a new generation of digital entrepreneurs ready to meet the challenges of the future.

## **DISCUSSION**

The integration of AI into e-entrepreneurship education offers profound opportunities to modernize and enhance entrepreneurial training. As Wei (2023) and Murtaza et al. (2022) have noted, AI technologies provide a personalized, scalable, and adaptive learning environment that can support the diverse needs of students in the rapidly evolving field of entrepreneurship. The personalization of learning, particularly through AI-powered adaptive instruction, allows students to learn at their own pace since they receive tailored feedback and guidance based on their individual progress and challenges (Labyad et al., 2024). This personalized approach addresses key gaps in traditional education, where one-size-fits-all curricula often fail to cater to the wide range of learner profiles in entrepreneurship education.

The findings of the thematic analysis align with the literature, emphasizing how AI tools, such as gamification and AI-powered business simulations, enhance engagement and promote experiential learning. Informants reflected on how AI's ability to simulate real-world business challenges allowed them to test entrepreneurial strategies without the risks associated with starting a real business (Zhao, 2021). By engaging with AI in a controlled environment, students can build practical skills and refine their decision-making abilities in ways that traditional education methods cannot replicate.

However, despite the clear advantages, the integration of AI into e-entrepreneurship education is fraught with challenges. The literature and informant responses consistently highlighted technical barriers, including insufficient digital infrastructure in many educational institutions (Aghaziarati et al.,



2023). These infrastructural deficits, particularly in developing regions, limit institutions' ability to implement AI-driven tools fully, since they require substantial computing power and stable internet connectivity. Furthermore, the complexity of AI systems poses challenges for both students and educators who may not be technically equipped to use these technologies. As noted by Geddam et al. (2024), the ease of use and accessibility of AI systems significantly impact their adoption, and their study highlights the need for professional development to help educators integrate AI tools effectively into their pedagogy.

Ethical issues related to data privacy and algorithmic bias remain significant barriers to the widespread adoption of AI in educational settings. As highlighted by informant 19, concerns about the collection and usage of personal data, alongside the potential for AI systems to perpetuate biases, underscore the importance of clear ethical guidelines and transparency in AI design. Studies by Ambarita and Nurahmatullah (2024) and Ouyang et al. (2023) further stressed the need for diverse datasets and unbiased algorithms to ensure that AI tools are equitable and serve the needs of all students. Institutions must prioritize the creation of ethical frameworks that not only protect user privacy but also mitigate the risks of perpetuating biases through AI systems.

The findings also suggest that, while AI tools can improve learning outcomes, student adaptability remains a critical factor in their successful implementation. Many students, particularly those with limited exposure to digital tools, may face challenges in adopting AI-based learning platforms. Educators play a crucial role in easing this transition by guiding students through AI tools and explaining their benefits, as suggested by educator 4. Without adequate support and gradual integration of AI tools, resistance from students can hinder the effectiveness of these technologies.

## IMPLICATIONS AND FUTURE RESEARCH

The findings of this study underscore the transformative potential of AI in e-entrepreneurship education. For educational institutions, particularly in resource-constrained regions, prioritizing investments in digital infrastructure is paramount. Access to reliable internet, modern computing devices, and cloud systems is essential to ensure that AI tools can be implemented effectively. These resources not only enable equitable access to technology but also facilitate the personalized learning environments that AI can offer, fostering entrepreneurial skills development among students from diverse backgrounds. In parallel, there is a compelling need for continuous professional development for educators. Training initiatives must focus on equipping educators with the knowledge and skills to integrate AI effectively into their teaching practices. This includes both technical training on AI tools and pedagogical strategies that leverage these tools to enhance learning outcomes. Well-prepared educators are central to realizing the full potential of AI in the classroom, ensuring that students derive maximum benefit from these innovations.

Moreover, fostering AI literacy among students is crucial to bridging the digital skills gap observed in many educational settings. Institutions should incorporate AI education into the curriculum, not only to teach students about the tools themselves but also to enhance their ability to use AI in entrepreneurial contexts. This would better prepare students for the digital demands of the modern business landscape, ensuring that they possess the necessary skills to succeed in an AI-driven economy. Ethical considerations are another key area of focus. As AI tools become more integrated into educational processes, institutions must develop frameworks to address concerns, such as data privacy, algorithmic bias, and the transparency of AI-driven decision-making. Establishing clear ethical guidelines is essential to fostering trust and ensuring that AI technologies are used responsibly and equitably in education.

Looking forward, future research should explore the long-term effects of AI integration on entrepreneurial education, particularly in terms of its impact on students' entrepreneurial competencies and outcomes. Longitudinal studies could provide valuable insights into how AI influences skill development over time. Comparative research across different regions and

educational settings could further illuminate context-specific challenges and offer practical recommendations for overcoming barriers to AI adoption. Additionally, future investigations should assess the student experience with AI-driven learning platforms to understand better how these tools affect their engagement, motivation, and learning outcomes. A more holistic understanding of students' interactions with AI tools could inform the design of more effective and inclusive educational programs.

Finally, scaling AI tools for inclusive entrepreneurship should be a priority for both researchers and policymakers. It is critical to design AI tools that cater to entrepreneurs from diverse backgrounds, particularly those in underrepresented and underserved areas. Making AI-driven education accessible to a broader population will be crucial for fostering inclusive innovation and entrepreneurship, particularly in emerging markets. While AI holds immense potential for revolutionizing e-entrepreneurship education, its successful integration depends on addressing infrastructure, training, and ethical challenges. By tackling these barriers, institutions can create dynamic, inclusive, and effective learning environments that better equip students for the demands of a rapidly evolving entrepreneurial landscape.

## **CONCLUSION**

The integration of AI into e-entrepreneurship education presents a transformative opportunity to reshape how entrepreneurial skills are taught and applied. This study highlights the potential of AI to enhance personalized learning, accelerate skill development, and increase student engagement in e-entrepreneurship programs. First, the findings show that AI-driven platforms can significantly enhance personalized learning by adapting content to individual student needs. These platforms, through AI-powered simulations and gamified tools, engage students in real-world entrepreneurial challenges, allowing them to improve essential skills, such as problem-solving and decision-making. This personalized, adaptive approach not only enhances student engagement but also promotes active learning, improving both retention and real-world application of entrepreneurial skills. Second, the study reveals key factors influencing the adoption of AI in educational contexts, as explored through the DoI framework. AI's relative advantage—such as automating tasks and personalizing learning—makes it an attractive tool for educators. However, challenges, such as compatibility with existing curricula, the complexity of AI systems, and the lack of technological proficiency among educators, create significant barriers to adoption. Addressing these issues through targeted training programs, simplified AI interfaces, and opportunities for trialability could ease adoption and lead to visible improvements in learning outcomes. Third, the study identifies practical challenges in integrating AI into e-entrepreneurship education, such as technical limitations, insufficient digital literacy among educators, and ethical concerns, such as data privacy and algorithmic bias. Institutions, especially in developing regions, face additional barriers, such as outdated infrastructure and limited resources. Moreover, ethical concerns related to AI usage—such as data security and fairness—highlight the need for robust protocols to ensure responsible and equitable implementation of AI tools.

In conclusion, AI has the potential to revolutionize e-entrepreneurship education by fostering personalized, dynamic learning environments. While the DoI framework provides valuable insights into the factors driving AI adoption, addressing practical challenges, such as infrastructure limitations, educator readiness, and ethical considerations, is essential. By overcoming these barriers, institutions can harness AI's full potential, creating more effective, inclusive, and engaging learning experiences for aspiring entrepreneurs.

## **LIMITATIONS OF THE STUDY**

While this study provides valuable insights into the integration of AI in e-entrepreneurship education, several limitations must be acknowledged. The geographical scope is limited, focusing primarily on institutions from select regions, which may not fully capture the diverse challenges and

opportunities that institutions face in other parts of the world, particularly in underrepresented areas. As a result, the findings may not be universally applicable. Additionally, the sample size, although adequate for the study's objectives, remains relatively small, potentially overlooking the varied experiences of different educators and institutions. A broader and more diverse sample would offer a more comprehensive understanding of AI integration in e-entrepreneurship education. The rapid pace of AI advancements is another limitation, as the findings of this study may become outdated with the introduction of new tools and methodologies. As AI technologies continue to evolve, so will the challenges and opportunities they present, which may differ from those identified here. Lastly, the reliance on self-reported data is also a limitation, as interviews and FGDs are subject to participant biases, which may affect the objectivity of the responses. Future research could mitigate this by employing additional data collection methods, such as observational studies or performance-based evaluations.

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